

# Surround Sound V2: Progress Report on Deep Learning Upgrades for Audio Scene Classification

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## Abstract

Surround Sound V2 extends a previously completed dual-head convolutional neural network (CNN) audio classification system with deep learning-specific architectural and training improvements. The original system, developed for CSS 581 (Machine Learning), achieved macro-F1 scores of approximately 0.50 for 8-class environment classification and 0.39 for 62-label sound event detection using log-mel spectrogram CNNs trained on AudioSet and FSD50K. This progress report describes the transition from that baseline toward a more expressive deep learning architecture. Planned and in-progress upgrades include Squeeze-and-Excitation (SE) channel attention blocks, residual skip connections, focal loss for long-tail class imbalance, and SpecAugment-based data augmentation. A third classification head for human speech detection is also under design. The unified preprocessing pipeline and real-time Streamlit inference demo from V1 remain intact as a stable foundation for these enhancements.

## 1 Introduction

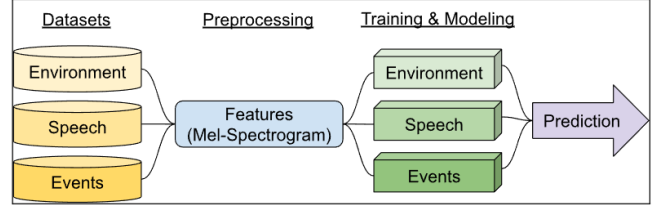
Surround Sound is a real-time audio scene understanding system that classifies both background environments and fine-grained sound events from microphone input. The system was originally developed for CSS 581 (Machine Learning) and demonstrated the viability of a dual-head CNN pipeline operating on log-mel spectrogram representations. V1 produced moderate performance: macro-F1 of approximately 0.50 for the 8-class environment classifier and 0.39 for the 62-label multi-label event detector. Additionally, it successfully enabled a real-time Streamlit demo with optional large language model (LLM) scene summarization [12].

Surround Sound V2, developed for CSS 586 (Deep Learning), builds directly on this foundation. Rather than redesigning the system from scratch, V2 introduces continuous integration of deep learning upgrades motivated by specific failure modes identified in V1: class imbalance, acoustic overlap between similar categories, limited temporal modeling, and insufficient expressive capacity in the convolutional backbone. This report describes the current state of those upgrades, the architectural decisions under exploration, and the plan for the final paper.

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**Figure 1: Overview of the Surround Sound V2 system architecture, showing dataset inputs, preprocessing pipeline, dual-head CNN models, and real-time inference output.**

The project diverges from single-task benchmarks such as the DCASE Challenge [10] by maintaining a multi-head design capable of simultaneous environment and event interpretation, with a real-time user-facing inference pipeline rather than offline evaluation.

## 2 V1 Baseline and Failure Mode Analysis

The V1 system consisted of two lightweight CNN classifiers sharing a common preprocessing backbone. Each model operated on 128-band log-mel spectrograms computed from 16 kHz mono audio standardized to 10-second segments. The environment classifier used a 4-layer Conv2D stack ( $32 \rightarrow 64 \rightarrow 128 \rightarrow 128$  channels) with softmax output over 8 mutually exclusive scene labels. The event classifier used a deeper stack ( $64 \rightarrow 128 \rightarrow 256 \rightarrow 256$  channels) with sigmoid output over 62 canonical sound event labels derived from FSD50K [1]. Both models used AdaptiveAvgPool2d as a global average pooling operation before the fully connected output layer.

Training for V1 used Adam optimization with cosine annealing, mixed-precision (`torch.cuda.amp`), and positive class reweighting (`BCEWithLogitsLoss` with `pos_weight` for events; weighted `CrossEntropyLoss` for environment). A Streamlit interface provided live audio capture, dual-head inference, and LLM-backed scene summarization. Aggregate results are summarized in Table 1.

**Table 1: V1 Baseline Performance**

| Model                 | Macro-F1 | Weighted-F1 |
|-----------------------|----------|-------------|
| Environment (8-class) | 0.50     | 0.45        |
| Events (62-label)     | 0.39     | 0.54        |

Post-training analysis identified four primary failure modes that directly motivate the V2 design:

**Class imbalance.** FSD50K’s data distribution caused the event classifier to severely underperform on rare classes (e.g., Scissors

F1 = 0.15, Aircraft F1 = 0.15), despite positive class reweighting. The modulating effect of standard cross-entropy is insufficient for extreme imbalance ratios.

**Acoustic overlap.** Acoustically similar categories — *indoor* vs. *other\_ambient*, or the *bird/bird-chirp/bird-call-song* family — were consistently confused. Lightweight CNNs without channel recalibration mechanisms lack the ability to selectively weight frequency bands per class.

**Limited temporal modeling.** Both models have limited temporal structure via global average pooling. Essentially, for events whose identity depends on temporal patterns (e.g., rhythmic percussion vs. ambient noise), the system’s ability to model how sound evolves over the 10-second window is difficult to portray.

**Weak LLM integration.** The V1 LLM summarization stage was a post-hoc add-on receiving raw classifier outputs, limiting the coherence of generated descriptions. V2 targets a more structured prompt interface.

### 3 Proposed V2 Architecture

V2 introduces four targeted architectural and training upgrades, each directly motivated by a V1 failure mode. The preprocessing pipeline and data index structure remain unchanged, ensuring that improvements are attributable to model and training changes rather than data artifacts.

#### 3.1 Squeeze-and-Excitation Channel Attention

To address acoustic overlap, V2 introduces Squeeze-and-Excitation (SE) blocks [6] into the deeper convolutional stages of both classifiers. SE blocks perform channel-wise recalibration: a global average pool compresses spatial information into a channel descriptor, which passes through a small bottleneck MLP and a sigmoid gate to produce per-channel attention weights. This allows the network to learn which mel-frequency bands are most discriminative for each class — for example, suppressing low-frequency background energy when classifying bird vocalizations. SE blocks are inserted after the third and fourth convolutional blocks in each model, where feature maps are most semantically abstract. A reduction ratio of 8 keeps parameter overhead minimal.

#### 3.2 Residual Skip Connections

The V1 event classifier’s  $256 \rightarrow 256$  convolutional block is a natural location for a residual skip connection [4]. A  $1 \times 1$  convolution serves as the shortcut projection to match channel dimensions. This addresses gradient flow issues in deeper training runs and encourages the network to learn incremental feature refinements rather than full transformations at each layer, which is particularly beneficial for the 40-epoch training segment used in V1.

#### 3.3 Focal Loss for Long-Tail Event Detection

V1 used BCEWithLogitsLoss with positive class weighting to handle label imbalance. While this improved recall for underrepresented classes, it did not address the model’s tendency to concentrate learning on easy, frequently-occurring examples. V2 replaces this with focal loss [8], which multiplies the standard binary cross-entropy by a modulating factor  $(1 - p_t)^\gamma$  that down-weights well-classified examples and concentrates gradient signal on hard,

misclassified instances. With  $\gamma = 2.0$  and existing positive class weights retained, focal loss provides a principled mechanism for improving rare-class recall without requiring oversampling, which would compromise clip diversity in the training set.

### 3.4 SpecAugment Data Augmentation

V1 tested time and frequency masking but did not apply it consistently, citing limited gains relative to training time. V2 reintroduces SpecAugment [11] within the `Dataset.__getitem__` method, applying stochastic time masking (up to 30 frames) and frequency masking (up to 20 mel bands) during training only. Applying augmentation at the dataset level rather than at preprocessing time generates unique augmented variants per epoch rather than fixed copies, increasing effective data diversity without increasing storage footprint.

## 4 Planned Speech Classification Head

A third classification head for human speech detection is planned as a V2 extension, completing the three-pillar architecture originally described in V1. V1 intentionally excluded speech as a separate classification task; the event classifier captures some speech-adjacent labels (Speech, Shouting, Singing, Human Group) but does not model conversation as a distinct category in scene classification.

The speech head will operate on the same log-mel preprocessing pipeline as the environment and event classifiers, using a task-specific label set and network head. Dataset selection is ongoing; candidate sources include Mozilla Common Voice and LibriSpeech for clean read speech, and AudioSet’s speech-related subclasses for naturalistic recording conditions. The primary design challenge is defining a label space that complements rather than overlaps the existing event taxonomy. Identifying words and conversations is the scope of this model.

## 5 Preserved Infrastructure from V1

Several components from V1 remain structurally unchanged in V2:

**Preprocessing pipeline.** Both `environment_preprocess.py` and `events_preprocess.py` produce consistent 128-band log-mel spectrograms at 16 kHz, hop length 512,  $f_{\max} = 8$  kHz using `librosa` [9]. This shared feature space ensures that a single spectrogram computation feeds all classification heads in the real-time demo.

**Data index and metadata handling.** Data index files and JSON label maps are reused directly. Hardcoded Windows path is corrected in V2 to use `Path(__file__).resolve().parents[1]` for cross-platform compatibility.

**Real-time Streamlit demo.** The live inference pipeline (audio capture  $\rightarrow$  spectrogram  $\rightarrow$  dual-head classification  $\rightarrow$  LLM summarization) remains functional and will be extended to include the speech head once trained.

**Evaluation pipeline.** The `eval.py` script supporting per-class precision, recall, F1, and confusion matrix generation is reused without modification.

## 6 Known Limitations and V2 Mitigations

Three preprocessing limitations identified during V1 code review are acknowledged and partially addressed in V2:

**Single-label resolution.** The `map_to_balanced()` function in `environment_label.py` returns the first matching category and discards co-occurring labels, contributing to boundary confusion between *outdoor\_nature*, *water*, and *other\_ambient*.

**Peak normalization only.** Both preprocessing scripts normalize by peak amplitude rather than perceptual loudness (LUFS). This introduces variability in mel representation for clips with uneven dynamic range. LUFS normalization via `pyloudnorm` is a planned addition.

**AudioSet clip availability.** V1 achieved approximately 37% retrieval success (2,183 of 5,900 expected environment clips). V2 expands coverage by incorporating the unbalanced AudioSet subset, yielding 11,207 environment samples, and applies strict label filtering to reduce noise from co-occurring speech and music.

## 7 Plan for the Final Paper

The final paper will present the following completed components:

- (1) Trained V2 environment and event classifiers with SE attention, residual connections, and focal loss, with ablation results comparing each upgrade against the V1 baseline.
- (2) Per-class F1 analysis and updated confusion matrices, with explicit attention to V1 failure modes (acoustic overlap, rare-class recall).
- (3) Speech head design and results, if dataset selection and training are feasible within the project timeline.
- (4) Extended real-time demo integrating all active classification heads.
- (5) Discussion of architectural alternatives considered but not implemented (e.g., Audio Spectrogram Transformer [3], PANNs [7] as feature extractors), with justification for the chosen design. This was the only complaint given with the final paper for Surround Sound V1.

## 8 Conclusion

Surround Sound V2 extends a proven baseline rather than rebuilding from scratch. Each architectural upgrade (SE attention, residual connections, and focal loss) targets a specific weakness identified in V1, keeping the system grounded in evidence rather than complexity for its own sake. The preprocessing pipeline and real-time inference demo carry forward unchanged, so V2 progress can be measured cleanly against V1 results. The final paper will present a full comparison across both classifiers, and if time permits, a speech classification head that completes the three-pillar system originally envisioned at the project's outset.

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